

Development Environment for Embedded Systems

Raspberry RPi 2/3

A Development System Setup

- RPi is an embedded computer using ARM-based SoC

Note: USB-TTL is different to USB-RS232 in serial signal levels



Host Computer

Serial link (USB-TTL)
Ethernet
JTAG



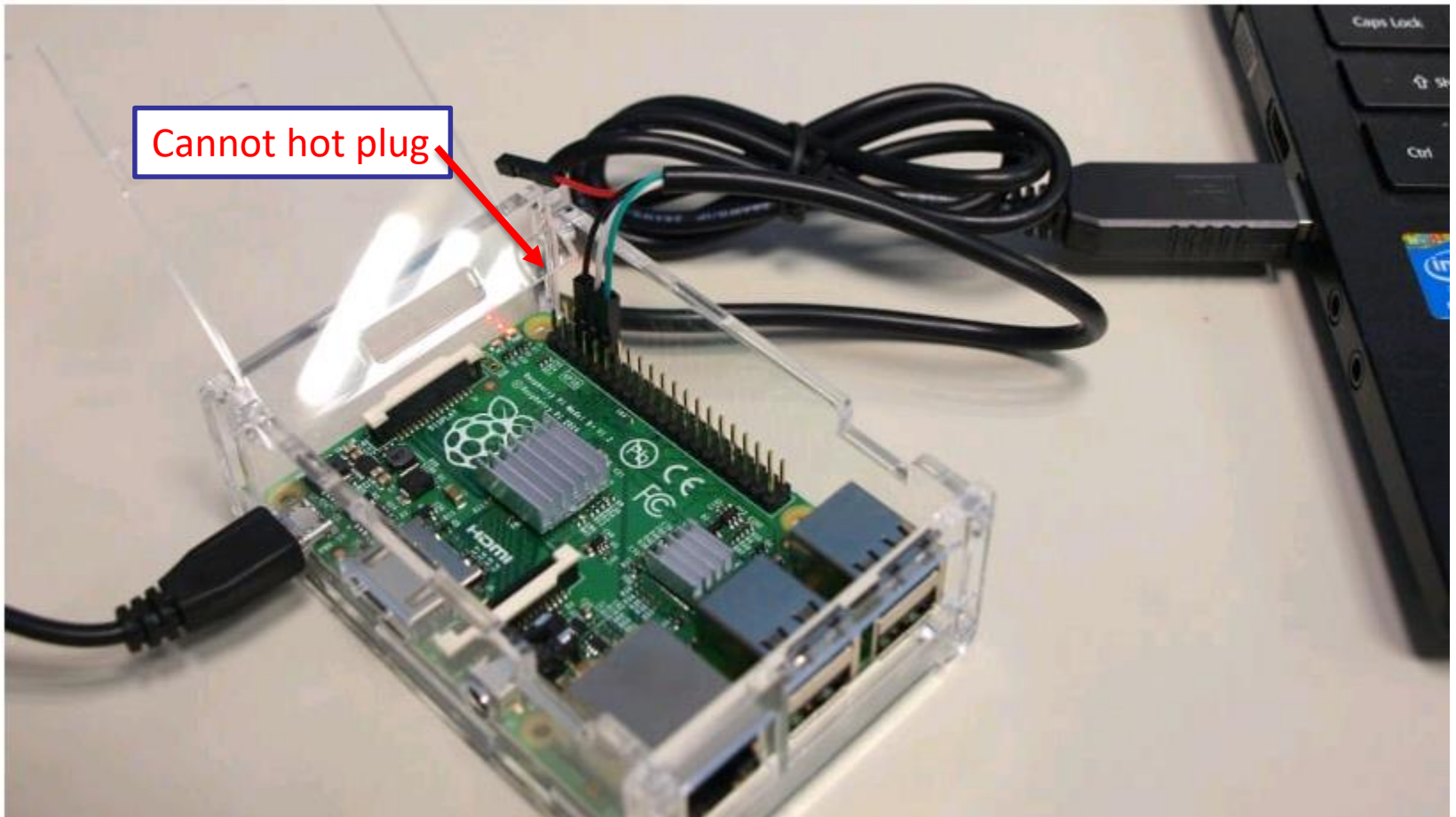
Target System
(Embedded System)

RPi

**ARM
SoC**

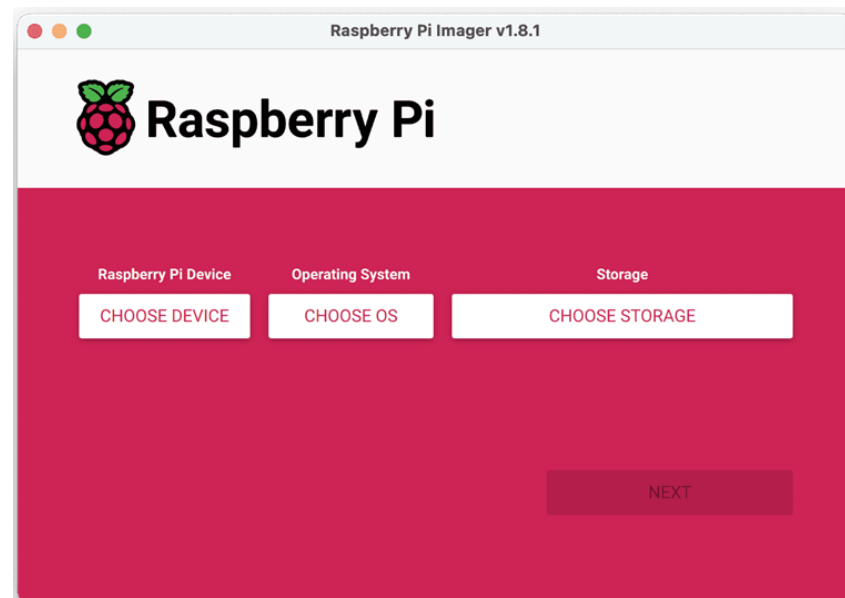
USB to TTL signal converter

- Host: use minicom to login Raspberry Pi
- Target: check **/boot/config.txt**
- Note: the signal pins cannot hot plug in/out



Installing Raspberry Pi OS to a microSD card

- Preparing a boot microSD card for your Raspberry Pi
- At the development host
 - Download and Install Raspberry Pi Imager
 - Raspberry Pi Imager is the quick and easy way to install Raspberry Pi OS to a microSD card, ready to use with your Raspberry Pi.

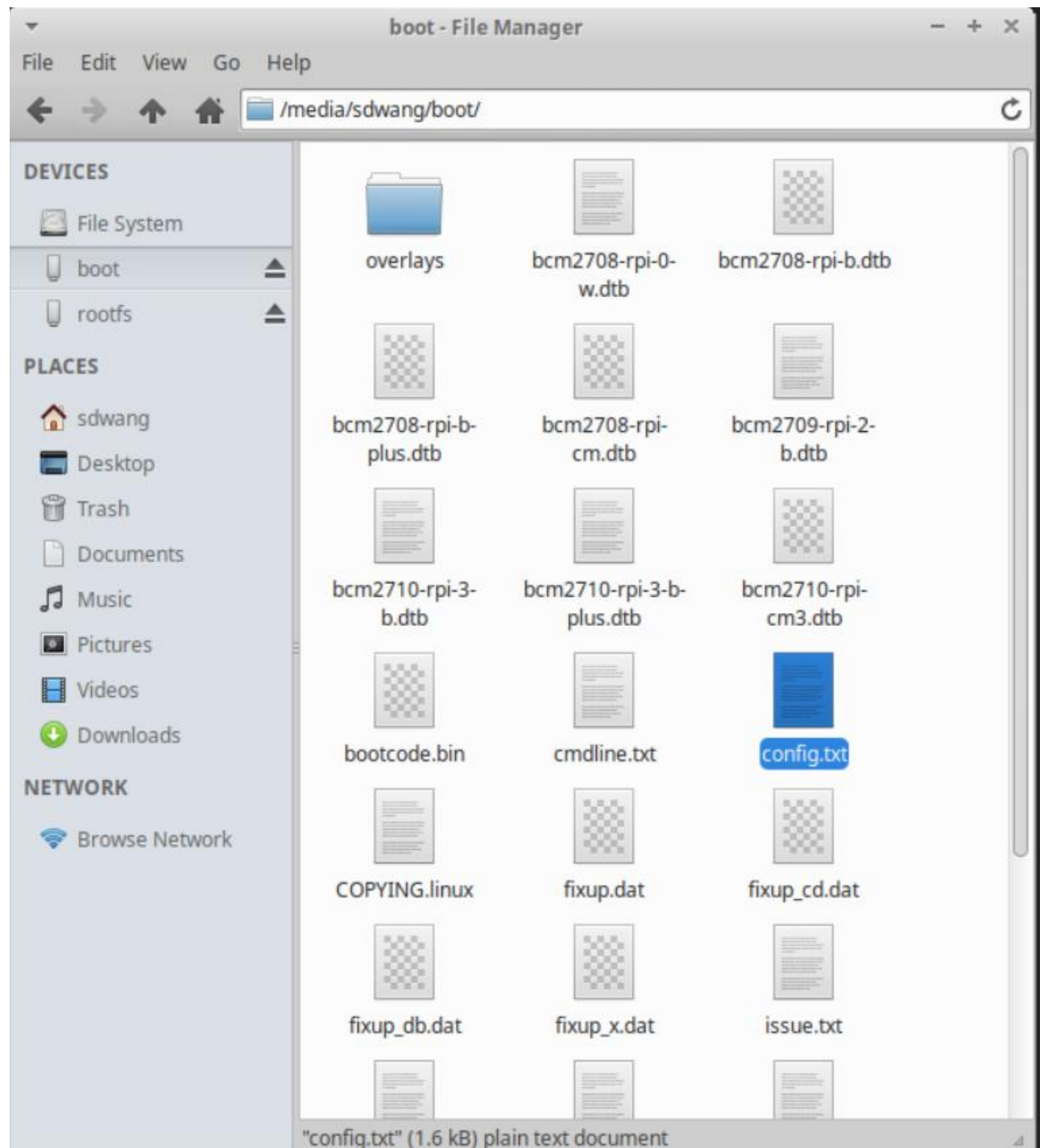


Raspberry Pi OS image file

- Include 2 partitions:
 - 1st partition: FAT format, firmware, kernel parameters, boot configurations
 - 2nd partition: ext4 format, as Linux root filesystem
- Check the microSD card after writing into it
- Check if config.txt at /boot of RPi boot microSD
 - Has the following line to communicate with the development host with UART serial port

```
enable_uart=1
```

- Check the SD to see the two partitions mounted in the development host



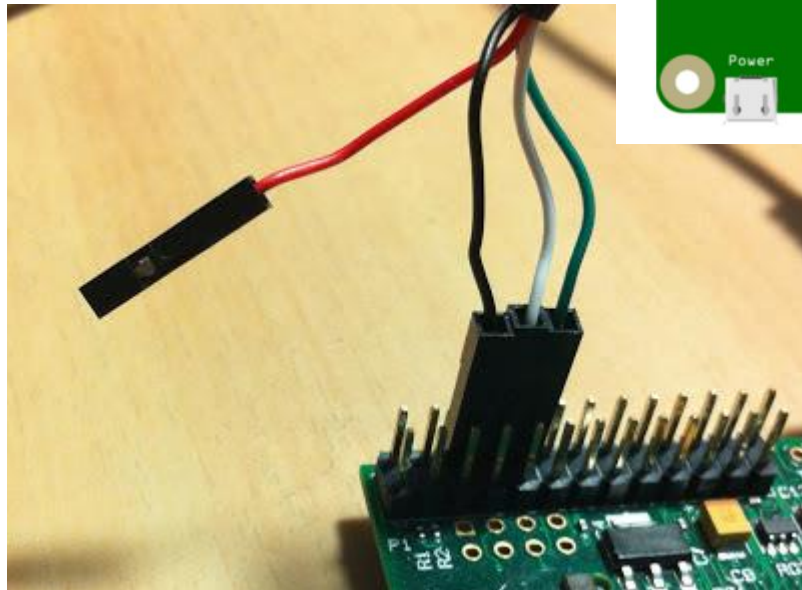
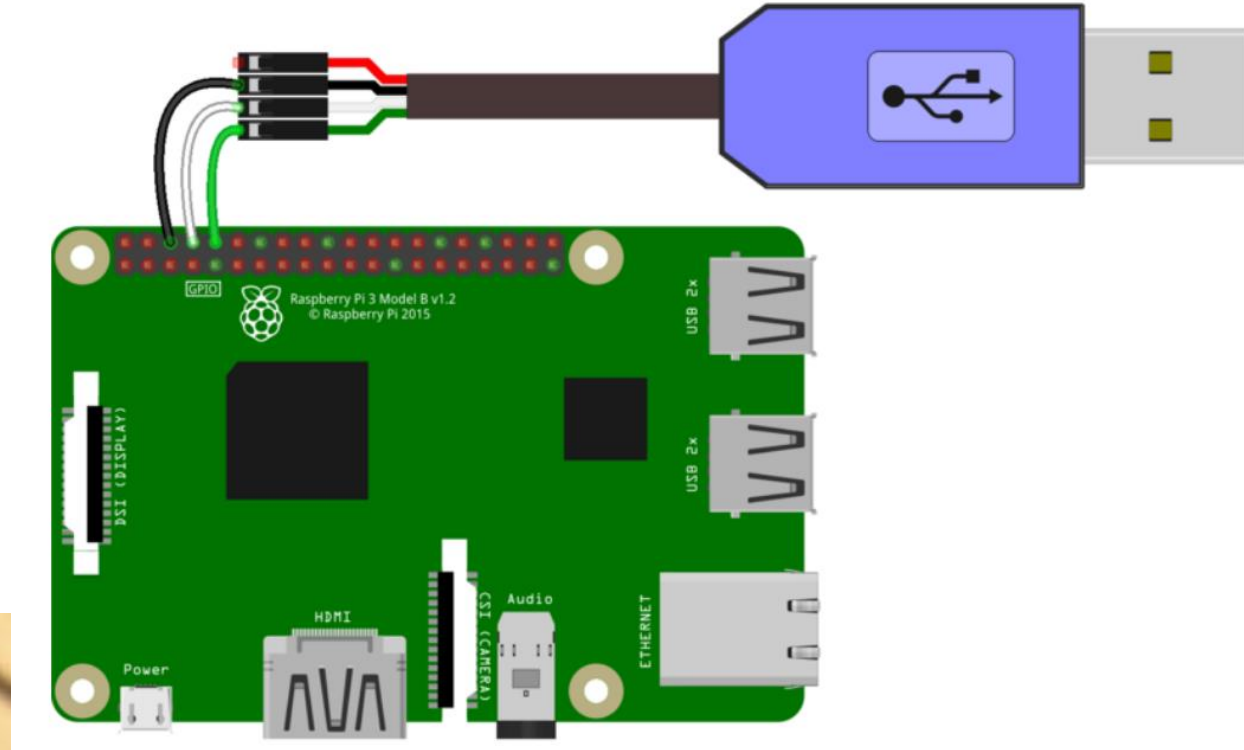
Verify what is on microSD card using command line

- (Host) `sudo fdisk -l /dev/sdb`
 - It should show at least two partitions, one is boot partition and another is the partition of a root file system.
1. Now the SD card should be bootable for your Raspberry pi
 2. Note: Some privileged command should be preceded with `sudo`

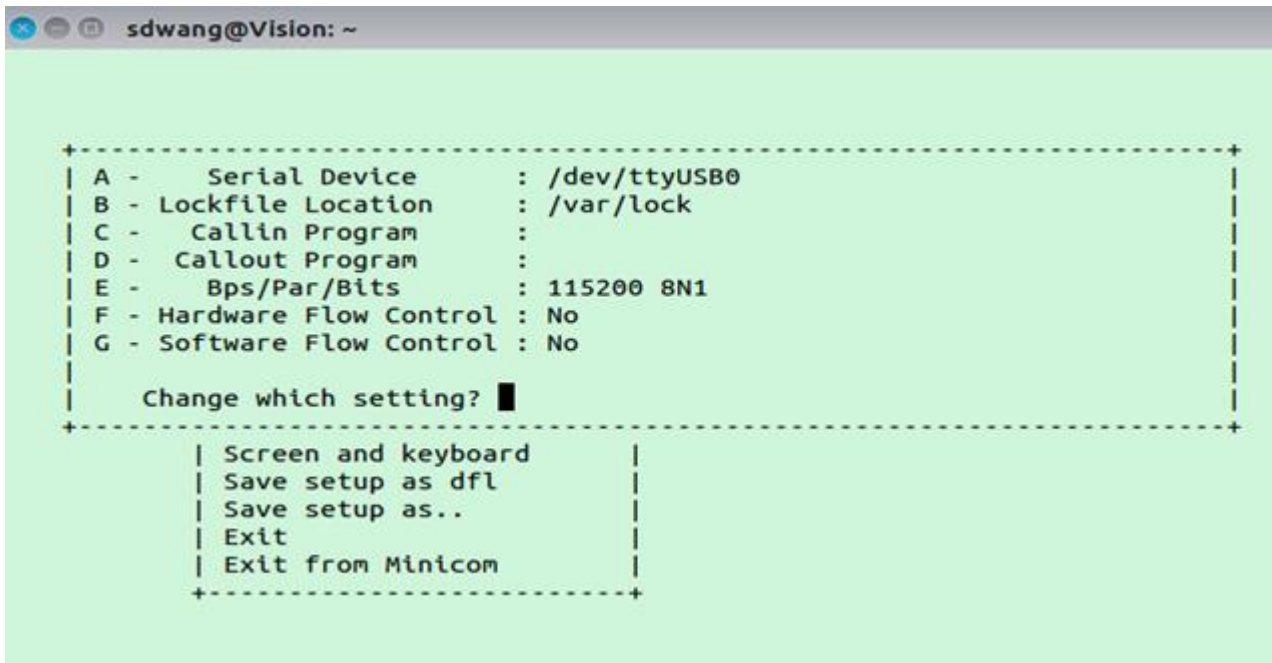
Use minicom to connect to RPi

- Check config.txt at /boot of RPi
 - (target) `cat /boot/config.txt`
`enable_uart=1`
- Use minicom to connect **USB-TTL** serial console to Raspberry Pi to see boot and console messages

USB-TTL



- # (run minicom for the first time, setting as below figure)
(Host) `sudo minicom -s`
- minicom
(Host) `sudo minicom`
- Or to enable line wrap
(target) `stty cols xx rows yy`
(target) `TERM=ansi`
where your minicom window is of size `xx` by `yy`



The screenshot shows a terminal window with the title bar "sdwang@Vision: ~". The terminal content displays the minicom configuration menu, which is enclosed in a dashed border. The menu lists several settings (A through G) and a prompt to change a setting. Below the main menu, there is a sub-menu with options like "Screen and keyboard", "Save setup as dfl", "Save setup as..", "Exit", and "Exit from Minicom".

```
sdwang@Vision: ~  
  
+-----+  
| A -   Serial Device       : /dev/ttyUSB0 |  
| B - Lockfile Location    : /var/lock     |  
| C -   Callin Program     :               |  
| D -   Callout Program    :               |  
| E -   Bps/Par/Bits       : 115200 8N1    |  
| F - Hardware Flow Control : No           |  
| G - Software Flow Control: No           |  
|                                         |  
| Change which setting? █                |  
+-----+  
|                                         |  
| | Screen and keyboard | |               |  
| | Save setup as dfl  | |               |  
| | Save setup as..   | |               |  
| | Exit               | |               |  
| | Exit from Minicom  | |               |  
| |                   | |               |  
+-----+
```

Booting Process

- GPU : mount SD card, load bootcode.bin
- bootcode.bin : initialize cache and memory, load start.elf
- start.elf : read **config.txt, cmdline.txt**,
- Initialize memory, load Linux Kernel, kernel.img
- CPU starts execution

(target) ls /boot

```
COPYING.linux
LICENCE.broadcom
LICENSE.oracle
bcm2708-rpi-b-plus.dtb
bcm2708-rpi-b.dtb
bcm2708-rpi-cm.dtb
bcm2709-rpi-2-b.dtb
bcm2710-rpi-3-b.dtb
bootcode.bin
cmdline.txt
config.txt
fixup.dat
fixup_cd.dat
fixup_db.dat
fixup_x.dat
issue.txt
kernel.img
kernel7.img
overlays
start.elf
start_cd.elf
start_db.elf
start_x.elf
```

raspi-config

- Linux System Configurations
- `raspi-config` is a shell script, open source

```
(target) sudo raspi-config
```

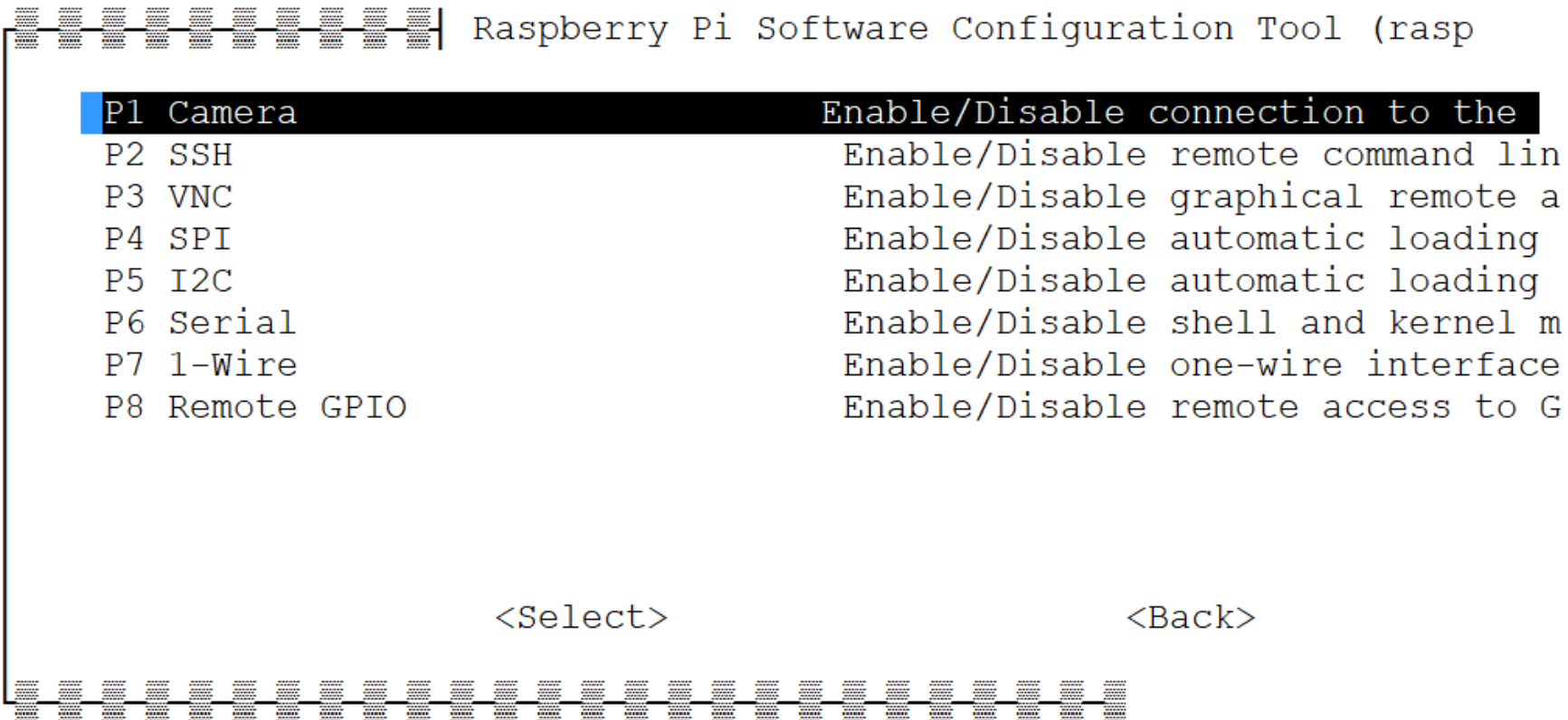
```
Raspberry Pi Software Configuration Tool (raspi-config)
```

1	Change User Password	Change password for the current user
2	Network Options	Configure network settings
3	Boot Options	Configure options for start-up
4	Localisation Options	Set up language and regional settings
5	Interfacing Options	Configure connections to peripheral devices
6	Overclock	Configure overclocking for your processor
7	Advanced Options	Configure advanced settings
8	Update	Update this tool to the latest version
9	About raspi-config	Information about this configuration tool

<select>

<Finish>

5. Interfacing Options



7. Advanced Options

Raspberry Pi Software Configuration Tool (raspi-config)

A1 Expand Filesystem

Ensures that all of the SD card space is used

A2 Overscan

You may need to configure overscan to see the full screen

A3 Memory Split

Change the amount of memory made available to the GPU

A4 Audio

Force audio out through HDMI or 3.5mm jack

A5 Resolution

Set a specific screen resolution

A6 Pixel Doubling

Enable/Disable 2x2 pixel mapping

A7 GL Driver

Enable/Disable experimental desktop mode

<Select>

<Back>

Reboot

- Command line interface
 - User: **pi** , password: **raspberry**
- Setup wifi
 - (target) `sudo raspi-config`

Setup wifi

- Using `raspi-config`

```
Raspberry Pi Software Configuration Tool (raspi-config)

1 Change User Password          Change password for the current user
2 Network Options               Configure network settings
3 Boot Options                  Configure options for start-up
4 Localisation Options          Set up language and regional settings
5 Interfacing Options           Configure connections to peripheral devices
6 Overclock                     Configure overclocking for your processor
7 Advanced Options              Configure advanced settings
8 Update                        Update this tool to the latest version
9 About raspi-config            Information about this configuration tool

<Select>                       <Finish>
```

Raspberry Pi Software Configuration Tool (rasp

- N1 Hostname Set the visible name for this Pi
- N2 Wi-fi Enter SSID and passphrase
- N3 Network interface names Enable/Disable predictable network

<Select>

<Back>

Enter SSID manually

Please enter SSID

<Ok>

<Cancel>

Setup wifi using command line

- Open the wpa-supPLICANT configuration file in nano or vi:

- (target)\$ `sudo nano /etc/wpa_supplicant/wpa_supplicant.conf`
 - Go to the bottom of the file and add the following:

```
network={
    ssid="testing"
    psk="testingPassword"
}
```

Or use `wpa_passphrase` to encode the PSK:

```
(target)$ wpa_passphrase "testing" | sudo tee -a
/etc/wpa_supplicant/wpa_supplicant.conf > /dev/null
```

After setup, reboot the RPi (or use commands: `sudo wpa_cli -i wlan0 reconfigure` and `sudo dhcpcd -n wlan0`)

Chang passwd and Shutdown Linux OS

- Change password
(target) `passwd`
- Shutdown Raspbian OS
 - (target) `sudo halt`
 - PWR LED still on
 - Green LED off
- Reboot
 - (target) `sudo reboot`

Update the Raspberry Pi OS on RPi

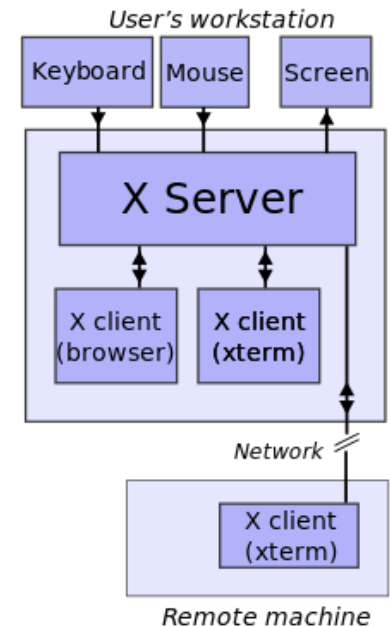
- Raspberry Pi OS is a Debian Linux distribution for RPi
 - Raspbian: 2019 version is Buster (Kernel 4.19, gcc 8.3)
 - 2024 version is Bookworm (Kernel 6.6, gcc 12.2)
 - Update the Raspberry Pi OS on RPi
- ```
(target) sudo apt update
```
- ```
(target) sudo apt -y dist-upgrade
```

Update the firmware

- For reference only, updating firmware is time consuming
- Firmware: some low-level hardware drivers that are not distributed along with Raspbian OS
 - (target) `sudo apt install rpi-update`
 - (target) `sudo rpi-update`
- Before Updating firmware
 - Backup important files/docs for safety
 - New firmware might have problems

RPi Remote Access

- Boot options (using `sudo raspi-config`)
 - Enable ssh server
- Command line
(host) `ssh <ip_RPi>`
- Remote GUI Access using X11 forwarding
 - At RPi target, install X client
 - xrdp will install rdp and X lib, ...
(target) `sudo apt install xrdp`
 - At Linux development host (**as X server**)
(host) `ssh -X <ip_RPi>`
 - At RPi target (**as X client**),
 - (target) `wget`
`https://timedotcom.files.wordpress.com/2017/04/ivanka-trump4.jpg` (舉例而已)
 - (target) `gpicview <a_jpg_file> &`
 - (target) `iceweasel &`
 - `gpicview` is a picture viewer and `iceweasel` is a web browser
- References
 - http://elinux.org/RPi_Remote_Access

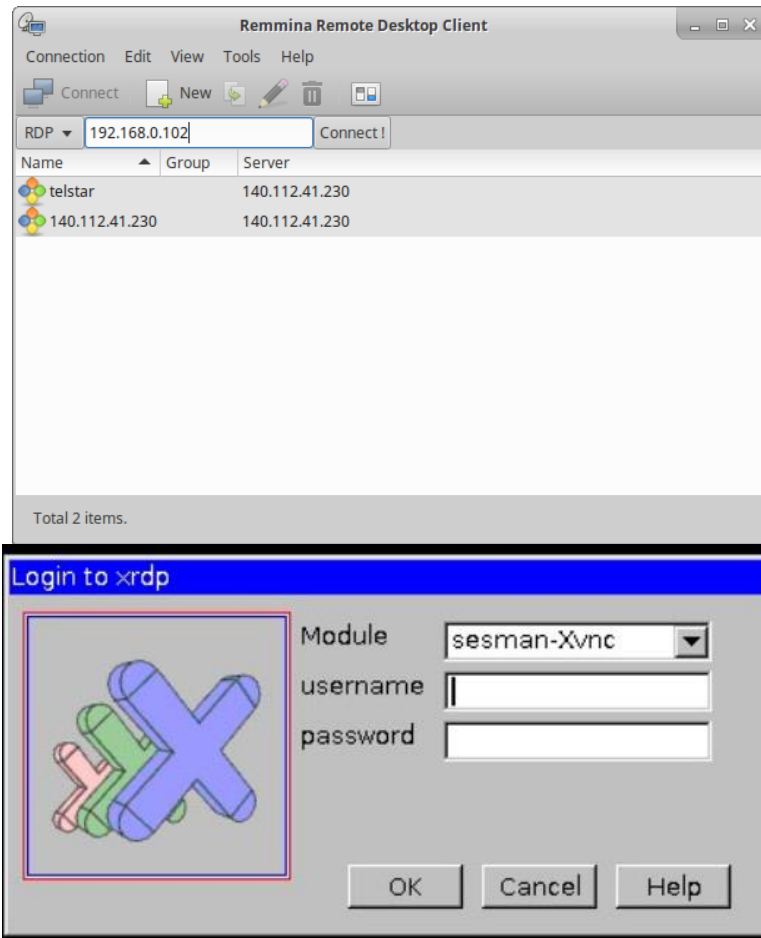


Remote desktop server

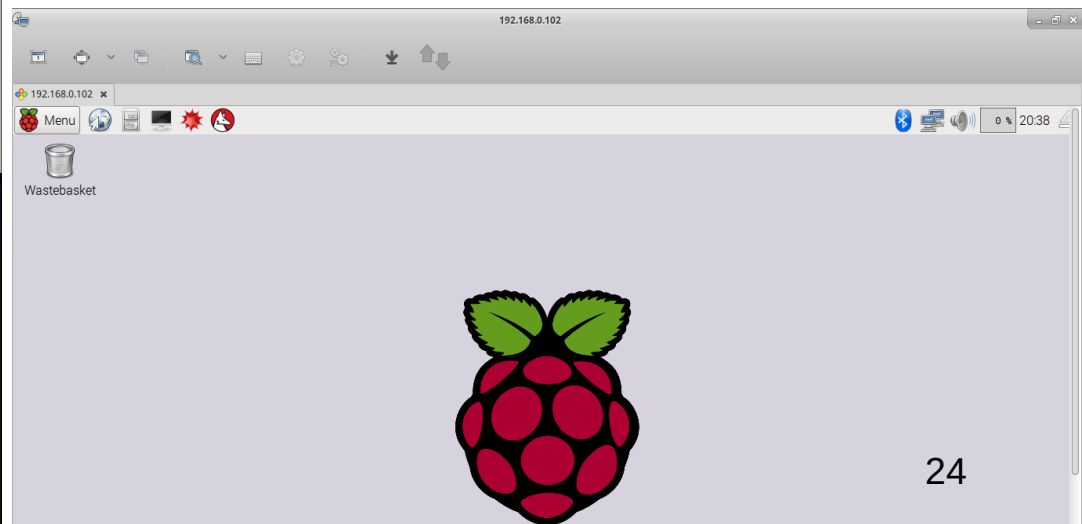
(target) `sudo apt install xrdp`

(target) `sudo apt install lxde (or raspberrypi-ui-mods)`

At the Linux host, use remmina to connect to remote desktop



Boot options (using `sudo raspi-config`) and choose boot into command line because we just want to login remotely



Pi 1, 2, and 3 Boards

Component / Board	Raspberry Pi 3 Model B	Raspberry Pi 2 Model B	Raspberry Pi Model B+
Processor Chipset	BCM 2837 64bit ARMv8 Cortex A53 Quad Core	BCM 2836 32bit ARMv7 Cortex A7 Quad Core	BCM2835 32bit ARMv11 Single Core
Processor Speed	1.2Ghz per core	900Mhz per core	700Mhz core
Max Power Draw	2.5A	1.8A	1.8A



RPi 2 and RPi 3 cpuinfo

cpuinfo - Raspberry Pi 3 Model B

```
processor      : 0
model name    : ARMv7 Processor rev 4 (v7l)
BogoMIPS     : 76.80
Features      : half thumb fastmult vfp edsp neon vfpv3 tls vfpv4 idiva idivt vfpd32
               lpae evtstrm crc32
CPU           : 0x41
implementer   :
CPU architecture: 7
CPU variant   : 0x0
CPU part      : 0xd03
CPU revision  : 4
<... and so on for each core...>
Hardware      : BCM2709
Revision      : a02082
Serial        : 0000000056163283
```

cpuinfo - Raspberry Pi 2 Model B

```
processor      : 0
model name    : ARMv7 Processor rev 5 (v7l)
BogoMIPS     : 57.60
Features      : half thumb fastmult vfp edsp neon vfpv3 tls vfpv4 idiva idivt
               vfpd32 lpae evtstrm
CPU           : 0x41
implementer   :
CPU architecture: 7
CPU variant   : 0x0
CPU part      : 0xc07
CPU revision  : 5
<... and so on for each core...>
Hardware      : BCM2709
Revision      : a01041
Serial        : 000000008e511ef0
```

(target) cat /proc/cpuinfo

Pi 2 and Pi 3 CPU Architectures

lscpu - Raspberry Pi 3 Model B	lscpu - Raspberry Pi 2 Model B
Architecture: armv7l	Architecture: armv7l
Byte Order: Little Endian	Byte Order: Little Endian
CPU(s): 4	CPU(s): 4
On-line CPU(s) list: 0-3	On-line CPU(s) list: 0-3
Thread(s) per core: 1	Thread(s) per core: 1
Core(s) per socket: 4	Core(s) per socket: 4
Socket(s): 1	Socket(s): 1
Model name: ARMv7 Processor rev 4 (v7l)	Model name: ARMv7 Processor rev 5 (v7l)
CPU max MHz: 1200.0000	CPU max MHz: 900.0000
CPU min MHz: 600.0000	CPU min MHz: 600.0000

Raspberry Pi 2 and 3 – I/O Ports

- 4 USB ports
- 40 GPIO pins
- Full HDMI port
- Ethernet port
- Combined 3.5mm audio jack and composite video
- Camera interface (CSI)
- Display interface (DSI)
- Micro SD card slot